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an, China, in December 2019. KI) in the setting of COVID-19 with worse clinical and renal outcomes¹⁻⁴. Possible mechanisms for kidney damage include direct and indirect viral effects⁵, interacting with common and known risk factors for AKI⁶.

The demonstration of viral RNA⁷⁻⁹ and live virus⁷ in kidney biopsies supports the hypothesis of a direct tropism of SARS-CoV-2 to the proximal tubule cells and podocytes, possibly via angiotensin-converting enzyme-2 receptor expressed in these cells¹⁰. SARS-CoV-2 is a cytopathic virus that can directly induce renal tubular injury during infection and replication, and the immune response can promote fibrosis and apoptosis⁵. Acute tubular necrosis (ATN)^{11,12}, thrombotic microangiopathy, and collapsing focal segmental glomerulosclerosis (FSGS) are reported histopathological findings⁶.

The indirect effects are those associated with critical illnesses, including hemodynamic instability, hypovolemia, hypoxia, cytokine storm, rhabdomyolysis, nephrotoxins exposure, endothelial dysfunction, coagulopathy, superimposed infections, and organ crosstalk^{5,13}. Clinical and autopsy reports indicate that SARS-CoV-2 infection induces a hypercoagulation state that increases the risk of blot clots, including pulmonary, splenic, and renal infarction⁶.

The reported incidence of AKI in patients with COVID-19, especially those who are hospitalized, varies depending upon the severity of disease in the group of patients analyzed. In two large observational studies with over 5,000 patients hospitalized with COVID-19, AKI was reported in 32 to 37% of patients^{14,15}. Approximately half of patients with AKI did not achieve complete recovery of kidney function by hospital discharge¹⁴. Independent predictors of AKI include age, race, gender, obesity, diabetes, hypertension, cardiovascular disease, low baseline estimated glomerular filtration rate (eGFR), higher interleukin-6 (IL-6) levels, or requirement of mechanical ventilation and vasoactive drugs^{14,15}.

There are limited data on AKI progression and long-term outcomes in critically ill patients with COVID-19. Most reports evaluated in-hospital outcomes^{13,14} and

Our goal was to describe the clinical and laboratorial profile and subsequent clinical course of a sample of critically ill patients with COVID-19 and AKI that required a nephrologist consultation in a tertiary hospital. The main outcomes were 90-day mortality and recovery of kidney function, as well as survival and estimated glomerular filtration rate at one year.

METHODS

DESIGN

We conducted a prospective cohort study with adult patients admitted to the Intensive Care Units (ICUs) of a tertiary hospital in Southern Brazil due to COVID-19 infection between May 1st, 2020, and April 30th, 2021.

AIM

Our primary goal was to evaluate short (90-day) and long-term (1-year) survival of a sample of critically ill patients with COVID-19 and AKI that required a nephrologist consultation (NC-AKI) in a tertiary hospital. The secondary goal was to describe the patients' clinical and laboratorial profile, AKI stages, and need for kidney replacement therapy (KRT) during hospitalization. Lastly, the recovery of kidney function or dialysis need after hospital discharge and at one-year follow-up was evaluated.

PATIENTS

Adult (≥ 18 years old) patients with COVID-19 admitted to the ICU with NC-AKI. Patients with end-stage renal disease in chronic KRT were excluded.

ETHICS

The study was approved by the local ethics committee (protocol number: 4237991). An informed consent form was obtained from patients or their next of kin.

DATA COLLECTION

Clinical and laboratory data were extracted from electronic medical records (EMR). Demographic data included age, gender, and body mass index (BMI). Comorbidities such as hypertension (HTN), obesity, diabetes mellitus (DM), malignancy, and coronary

was confirmed by quantitative reverse transcription polymerase chain reaction (RT-PCR). Baseline creatinine level was defined as the mean creatinine value between 7 and 365 days before hospitalization retrieved from the EMR at the hospital or through the patient's family. If that information was lacking, the minimum creatinine value during hospitalization was used as the baseline creatinine. AKI stages were defined according to KDIGO (Kidney Disease: Improving Global Outcomes) criteria as follow: stage 1 as an increase in serum creatinine of 1.5–1.9 times the baseline value or an increase in serum creatinine of 0.3 mg/dL within 48 hours; stage 2 as an increase of 2–2.9 times the baseline value; and stage 3 as an increase of at least 3 times the baseline value. Urine output was not used for the definition of AKI due to the scarce documentation of this variable. Positive albuminuria was defined as 2+ (or 1 g/dL) on urinalysis since lower values could be due to either ATN or acute interstitial nephritis (AIN). Frequency and modality of KRT – intermittent, continuous, or both – were reported. Peritoneal dialysis was not performed. Recovery of kidney function was independence of dialysis or serum creatinine within 50% of baseline according to Acute Dialysis Quality Initiative¹⁹.

OUTCOMES AND FOLLOW-UP

The primary outcomes within 90 days after COVID-19 infection included death, recovery of kidney function, and kidney function impairment with requirement of maintenance dialysis. Surviving patients had serum creatinine analysis at discharge and at 12 months. They were contacted by phone call and asked about their clinical status. Those who had missing values of serum creatinine were invited to collect new blood samples. Glomerular filtration rates were calculated using the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation at baseline, discharge, and 12 months.

STATISTICAL ANALYSIS

Continuous variables are presented as median with interquartile range (IQR) or mean and standard

between demographic data, clinical characteristics, and laboratory findings and death at 90 days and 1 year, χ^2 test, Fisher's exact test, and Mann-Whitney U test were used according to the nature and distribution of the variables. Variables with $P < 0.20$ were included in the multivariate analyses. Survival curves were constructed with Kaplan-Meier and compared by log-rank test. Cox proportional hazards regression analysis was used to determine hazard ratios at follow-ups. Creatinine values and eGFR at baseline, discharge, and one year after COVID-19 infection were compared with ANOVA for repeating variables and Sidak for multiple comparisons. The IBM SPSS® v. 25 was used for data analysis. Statistic tests were two-sided, and levels of significance were 0.05.

RESULTS

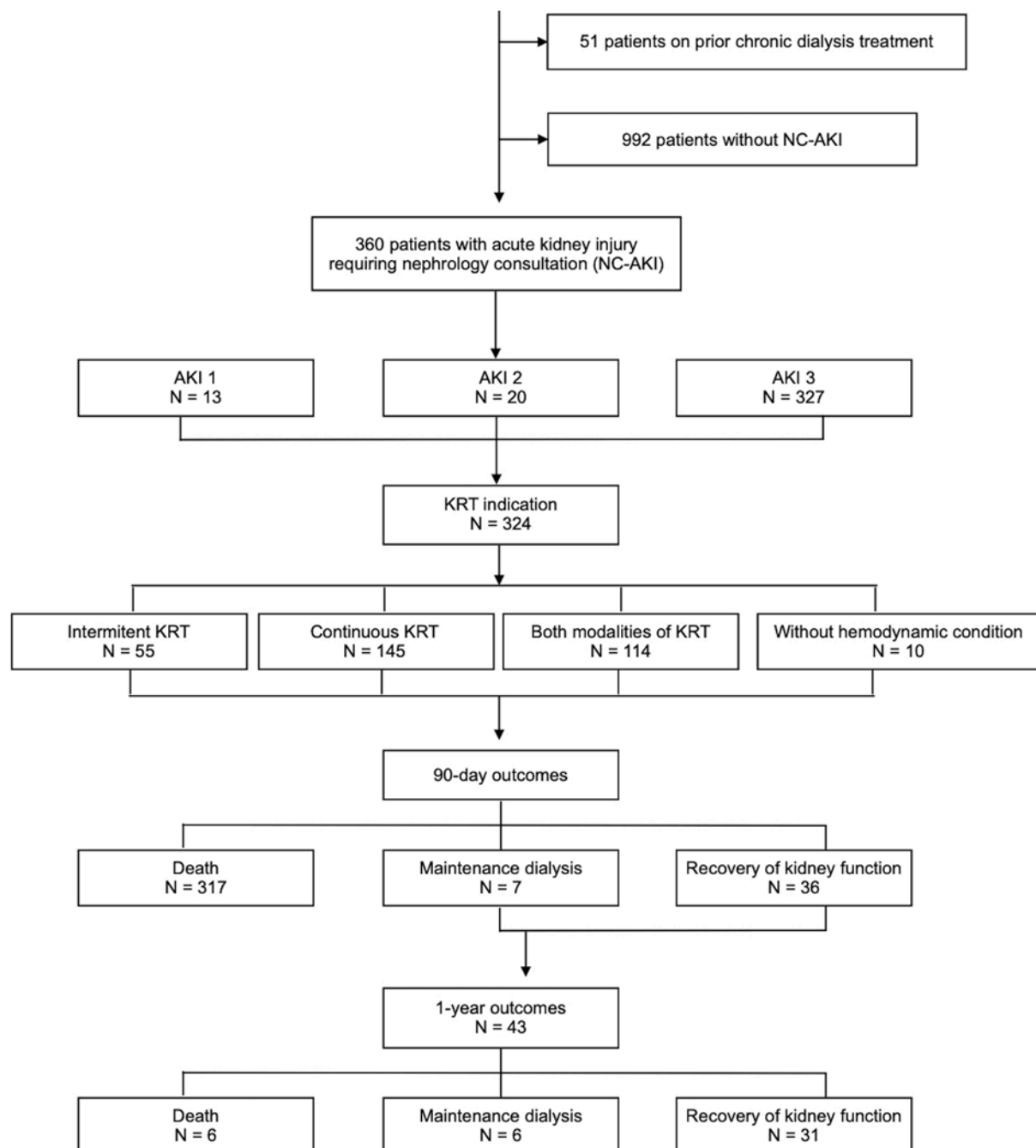


Figure 1. Flowchart of patients with COVID-19 hospitalized in the ICU during the study period. Patients with acute kidney injury (AKI) requiring nephrology consultation (NC-AKI) were included in the analysis. Outcomes were collected at 90 days and 1 year.

90-DAY SURVIVAL

In the first 90 days after COVID-19 diagnosis, 317 (88.1%) patients died, all during hospitalization.

Cumulative 90-day survival is presented in Figure 2.

	N = 360
ics	
	218 (60.6)
Age (years) (median, IQR)	66.0 (57.0–72.0)
BMI (kg/m ²) (median, IQR)	29.0 (25.0–33.2)
Baseline creatinine (mg/dL) (mean, SD)	1.34 (0.98)
Comorbidities (N,%)	
Hypertension	247 (68.6)
Diabetes mellitus	137 (38.1)
Obesity	142 (39.4)
Malignancy	64 (17.8)
Organ transplant recipients	59 (16.4)
Ischemic cardiopathy	61 (16.9)
Hospitalization factors	
Albuminuria (2 ⁺)	106 (29.3)
AKI Stage (N, %)	
Stage 1	13 (3.6)
Stage 2	20 (5.6)
Stage 3	327 (90.8)
KRT (N, %)	
Yes	314 (87.2)
No (without hemodynamic conditions)	10 (2.8)
No indication	36 (10.0)
Mechanical ventilation, (N, %)	352 (97.8)
Laboratorial data (median, IQR)	
CPK (U/L)	219.0 (89.0–658.0)
D-dimer (mg/L)	4.4 (1.8–4.4)
Ferritin (ng/mL)	1625 (848–2584)
Fibrinogen (mg/dL)	529 (410–679)
Procalcitonin (ng/mL)	1.3 (0.4–3.6)
Lactate (mmol/L)	3.7 (2.7–6.3)
C-reactive protein (mg/dL)	266.4 (214.1–332.8)
Troponin (pg/mL)	45.9 (11.4–300.7)

Abbreviations - BMI, body mass index; AKI, acute kidney injury; KRT, kidney replacement therapy; CPK, creatine phosphokinase; CRP, C reactive protein.

requirement of KRT ($p < 0.0001$), continuous KRT ($p < 0.0001$), AKI stage 3 ($p = 0.005$), malignancy ($p = 0.048$), mechanical ventilation ($p < 0.0001$), and superimposed bacterial infection ($p < 0.0001$) were significantly associated with death at 90 days (Table 2). The multivariate analysis for death at 90 days included variables with $p < 0.20$: age, gender, neoplasm, anticoagulation, mechanical ventilation, steroids, superimposed infection, dialysis, and C reactive protein. Table 3 shows the variables that remained associated with death at 90 days: older age [HR 1.015

90-DAY RENAL OUTCOMES

Forty-three (11.9%) patients survived beyond 90 days. At discharge, 7 (1.9%) remained on KRT and 36 (10%) had recovery of kidney function (RKF). There was an association of RKF at 90 days with no dialysis requirement ($p = 0.024$), anticoagulation ($p = 0.036$), and steroids use ($p = 0.022$). On multivariate analysis, only no KRT requirement remained significantly associated with RKF (HR 2.14, 95%CI 1.44–3.17, $p = 0.009$). Time to recovery of kidney function according to AKI stages is presented in Figure 4.

KRT requirement, KRT dependence, and death were grouped as major adverse kidney events (MAKE) for composite analysis. A multivariate analysis with age, gender, diabetes, hypertension, obesity, neoplasm, ischemic cardiopathy, steroids, anticoagulation, and baseline creatinine yielded a significant association with female gender (HR 1.256, 95%CI 1.01–1.56, $p = 0.041$), older age (HR 1.01, 95%CI 1.003–1.022, $p = 0.009$), and higher baseline creatinine (HR 1.14, 95%CI 1.03–1.27, $p = 0.009$).

ONE-YEAR FOLLOW-UP

The 43 patients that survived beyond 90 days were followed-up for 1 year: 6 (14%) died, 6 (14%) were receiving KRT (2 initiated and 4 remained), and 31 (72%) were free from dialysis. There were no significant associations of clinical variables with 1-year death.

Six (14%) patients that recovered kidney function had missing values of creatinine at one year. In the remaining 25 (58.1%) patients, the mean eGFR did not differ between baseline and discharge, but was

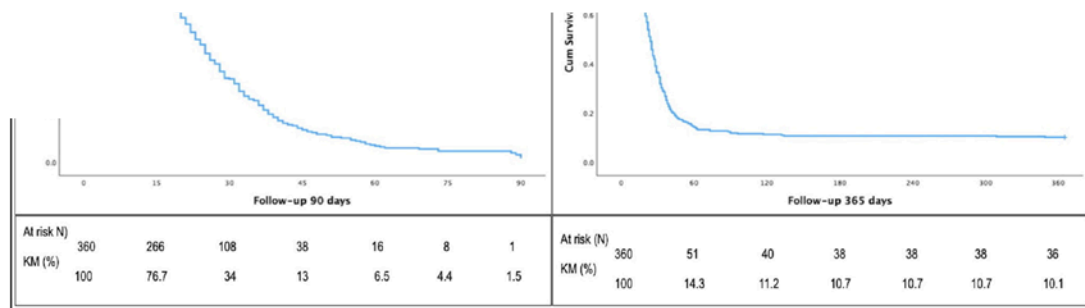


Figure 2. Left panel, 90-day survival of critically ill COVID-19 patients with acute kidney injury that required nephrology consultation. Right panel, 365-day survival of COVID-19 critically ill patients with acute kidney injury that required nephrology consultation.

TABLE 2 RISK FACTORS FOR DEATH AT 90-DAYS FOLLOW-UP OF CRITICALLY ILL COVID-19 PATIENTS WITH ACUTE KIDNEY INJURY THAT REQUIRED NEPHROLOGY CONSULTATION IN A TERTIARY HOSPITAL

	Survival (N = 43)	Death (N = 317)	P value
Male gender (%)	32 (74.4)	186 (58.7)	0.047
Age (mean, DP)	57.7 (13.2)	64.8 (12.1)	0.001
BMI (Median, IQR)	30.4 (34.3–25.4)	28.7 (32.8–25.0)	0.466
AKI stage (N, %)			
Stage 1	5 (11.6)	8 (2.5)	0.005
Stage 2	4 (9.3)	16 (5.0)	
Stage 3	34 (79.1)	293 (92.4)	
KRT (N, %)			
Yes	31 (72.1)	283 (89.3)	<0.001
No condition	0	10 (3.2)	
No indication	12 (27.9)	24 (7.6)	
KRT modality (N, %)			
Intermittent	15 (34.8)	40 (12.6)	<0.001
Continuous	2 (4.7)	143 (45.1)	
Both	14 (32.6)	100 (31.5)	
Obesity (N, %)	23 (53.5)	119 (42.8)	0.189
Hypertension (N, %)	28 (65.1)	219 (69.1)	0.603
Diabetes mellitus (N, %)	15 (34.9)	122 (38.5)	0.739
Malignancy (N, %)	3 (7.0)	61 (19.2)	0.048
Ischemic cardiopathy (N, %)	6 (14.6)	55 (17.4)	0.657
Organ transplant (N, %)	9 (20.9)	50 (36.3)	0.391
Mechanical ventilation (N, %)	37 (86.0)	314 (99.4)	<0.001
Anticoagulation	32 (74.4)	270 (85.4)	0.063
Superimposed infection	38 (88.4)	313 (99.7)	<0.001
Albuminuria (N, %)	15 (34.9)	91 (28.7)	0.347
CPK (U/L) (median, IQR)	141 (80–334)	227 (95–660)	0.160
D-dimer (mg/L) (median, IQR)	4.4 (2.1–4.4)	4.4 (1.7–4.4)	0.404
Ferritin (ng/mL) (mean, SD)	1643 (1746.7)	3350 (7256.7)	0.023
Fibrinogen (mg/dL) (mean, SD)	456.7 (244.7)	553.7 (192.6)	0.035
Lactate (mmol/dL) (mean, SD)	3 (1.1)	5.6 (4.1)	<0.001
Procalcitonin (ng/mL) (median, IQR)	0.8 (0.3–2.0)	1.4 (0.4–4.1)	0.119
CRP (mg/L) (median, IQR)	269.5 (203.8–311.8)	266.2 (217.3–335.1)	0.312
Troponin (pg/mL) (median, IQR)	29.1 (1.4–143.0)	54.3 (11.7–350.0)	0.179

	0.006	1.014	1.004–1.024	0.002	1.015	1.006–1.025
	0.028	1.290	1.028–1.619	0.014	1.325	1.059–1.659
	0.004			0.004		
No condition		2.962	1.406–6.239		3.017	1.435–6.342
No indication		1			1	

Note – Variables included in the model with no significant association on multivariate analysis: neoplasm, anticoagulation, mechanical ventilation, superimposed infection, steroid use, and C reactive protein.

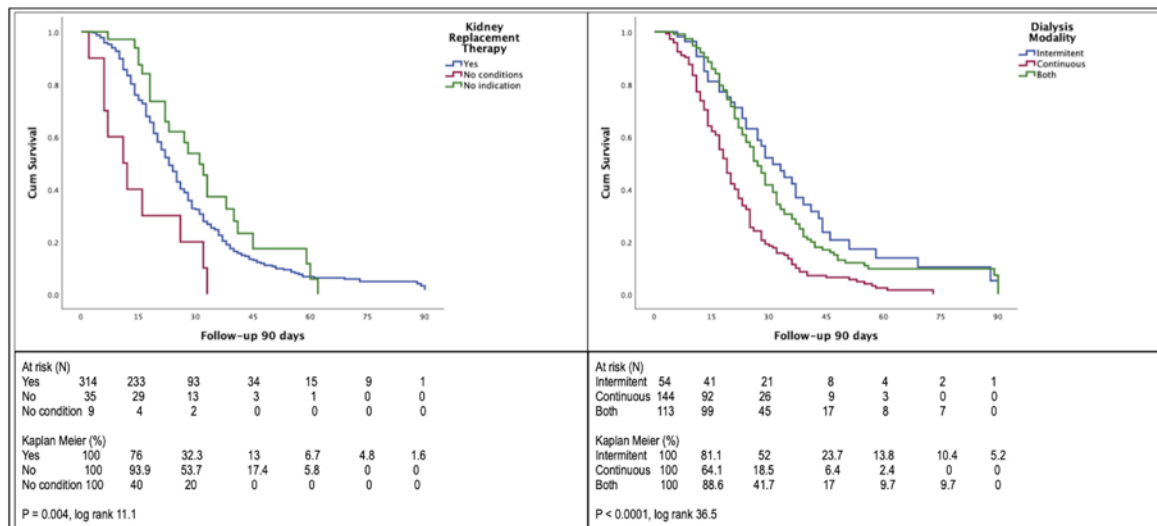
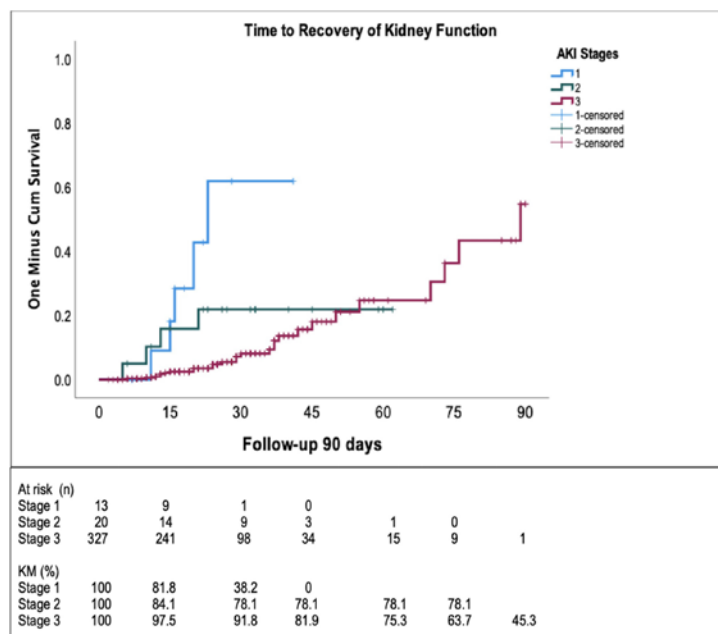


Figure 3. Left panel, 90-day survival of critically ill COVID-19 patients with acute kidney injury stratified by kidney replacement therapy requirement. Right panel, 90-day survival of critically ill COVID-19 patients with acute kidney injury that required kidney replacement therapy stratified by dialysis modality.



Notes – Stage 1: 5/13 (38.2%) patients had RKF in a median follow-up of 16.0 (13.0–21.5) days. Stage 2: 4/20 (20%) patients had RKF in a median follow-up of 11.5 (6.2–19.0) days. Stage 3: 27/327 (8.25%) patients had RKF in a median follow-up of 29.0 (15.0–45.0) days (log rank 30.6, p < 0.001).

G NEPHROLOGY CONSULTATION		
	Mean, SD	P value
	85.5 ± 23.6 ^a	0.003
Hospital discharge	81.7 ± 29.5 ^a	
One-year follow-up	65.9 ± 24.8 ^b	

Abbreviations - eGFR, estimated glomerular filtration rate (CKD-EPI).
Post-hoc Sidak test (p > 0.05).

significantly lower at 1-year follow-up (85.5 ± 23.6 vs. 81.7 ± 29.5 vs. 65.9 ± 24.8, p = 0.003) (Table 4).

DISCUSSION

In this prospective single-center cohort study we described critically ill COVID-19 patients with AKI that required nephrologist consultation (NC-AKI) in the ICUs of a tertiary hospital, representing 25% of ICU admissions. Most patients were male, in their sixth decade of life, with hypertension. The majority of patients presented stage 3 AKI and required kidney replacement therapy (KRT). Mortality within 90 days of follow-up was high. Amongst the surviving patients, most had recovery of kidney function at one-year follow-up, although with lower eGFR compared to baseline.

Our NC-AKI frequency was similar to other studies (22–36%)^{1,4,14,15} and lower than other Brazilian reports (55%)^{20,21}. The reported incidence of AKI depends on the severity of disease, the need for hospitalization, the criteria adopted to define AKI, and the study design, and has changed over time as more data on the extra-pulmonary implications of this new disease have become available²².

The pathogenic mechanism by which SARS-CoV-2 affects the kidney is likely multifactorial, involving both viral effects⁵ and known risk factors for AKI⁶. The most frequent kidney biopsy finding in COVID-19 patients is acute tubular injury^{11,12}, reflecting the numerous indirect causes of AKI seen in critical illness. These include hemodynamic instability, acute respiratory distress syndrome, exposure to nephrotoxins, hypoxia, cytokine storm, rhabdomyolysis, and secondary infections^{5,13}. Additionally, detection of SARS-CoV-2 RNA^{7–9} and live virus⁷ in the kidney supports the hypothesis

er KFR as cessation of KRT, –92%^{16,17,25,27} among surviving reported that 74% of their patients had RKF, defined as a decline of 33% from the peak admission serum creatinine level.

Although short-term outcomes of patients with AKI are widely reported^{29–32}, data on long-term follow up of COVID-19 survivors with in-hospital AKI are more limited^{4,18,33}. The presence of lung fibrosis observed after pulmonary infections from other coronavirus strains³⁴ has raised concerns that COVID-19-associated AKI might induce tubulointerstitial fibrosis³⁵, a potential pathway for progression from AKI to CKD^{36,37}. The long-term decline in kidney function after AKI has been described in specific scenarios, such as patients discharged after ICU admission or those who underwent coronary angiography^{38,39}. Understanding how COVID-19-related AKI impacts the long-term trajectory of kidney function can help clinicians tailor specific and thorough monitoring plans for managing kidney disease for recovering patients.

At one year follow up, 14% of survivors remained on KRT, another 14% died, and 72% were free from dialysis. Renal function analysis of this later subgroup showed that eGFR did not differ from baseline to hospital discharge. We have to consider the possibility of lower eGFR accuracy due to loss of muscular mass in the majority of patients. However, median eGFR one-year post-diagnosis was lower than both baseline and discharge levels, indicating partial recovery of kidney function.

The strengths of our study include its prospective nature and the assessment of patient survival and renal outcomes up to one year after COVID-19 infection. The limitations include the single-center nature with a population that required nephrological consultation, therefore limiting the generalizability of findings. Also, we recognize that the lack of measurement of urinary output for AKI staging, the use of qualitative albuminuria, and the missing creatinine measurements prior to hospitalization to define baseline eGFR are shortfalls in our analysis. Furthermore, some patients with AKI stage 1 and stage 2 may have been omitted

with NC-AKI presented a high frequency of AKI stage 3 and KRT requirement, with a high 90-day mortality. Surviving patients had high rates of recovery of kidney function, with lower eGFR at one-year follow-up compared to baseline.

AUTHORS' CONTRIBUTIONS

All authors contributed substantially to the development of this manuscript. JAMA and EK led the conceptualization, curation, analysis, data administration, supervision, writing and review of the manuscript. RP, ER, MADA contributed to methodology, investigation and data curation. CB contributed to data curation, methodology and formal analysis. FN and GB contributed in conceptualization, methodology and review of the manuscript. GM contributed to data analysis, writing and review of the manuscript. All authors approved the final version.

CONFLICT OF INTEREST

The authors do not have any conflicts of interest to declare.

REFERENCES

